OccPoIs: Points of Interest based on Neural Network's Key Recovery in Side-Channel Analysis through Occlusion

Trevor Yap, Shivam Bhasin, Stjepan Picek

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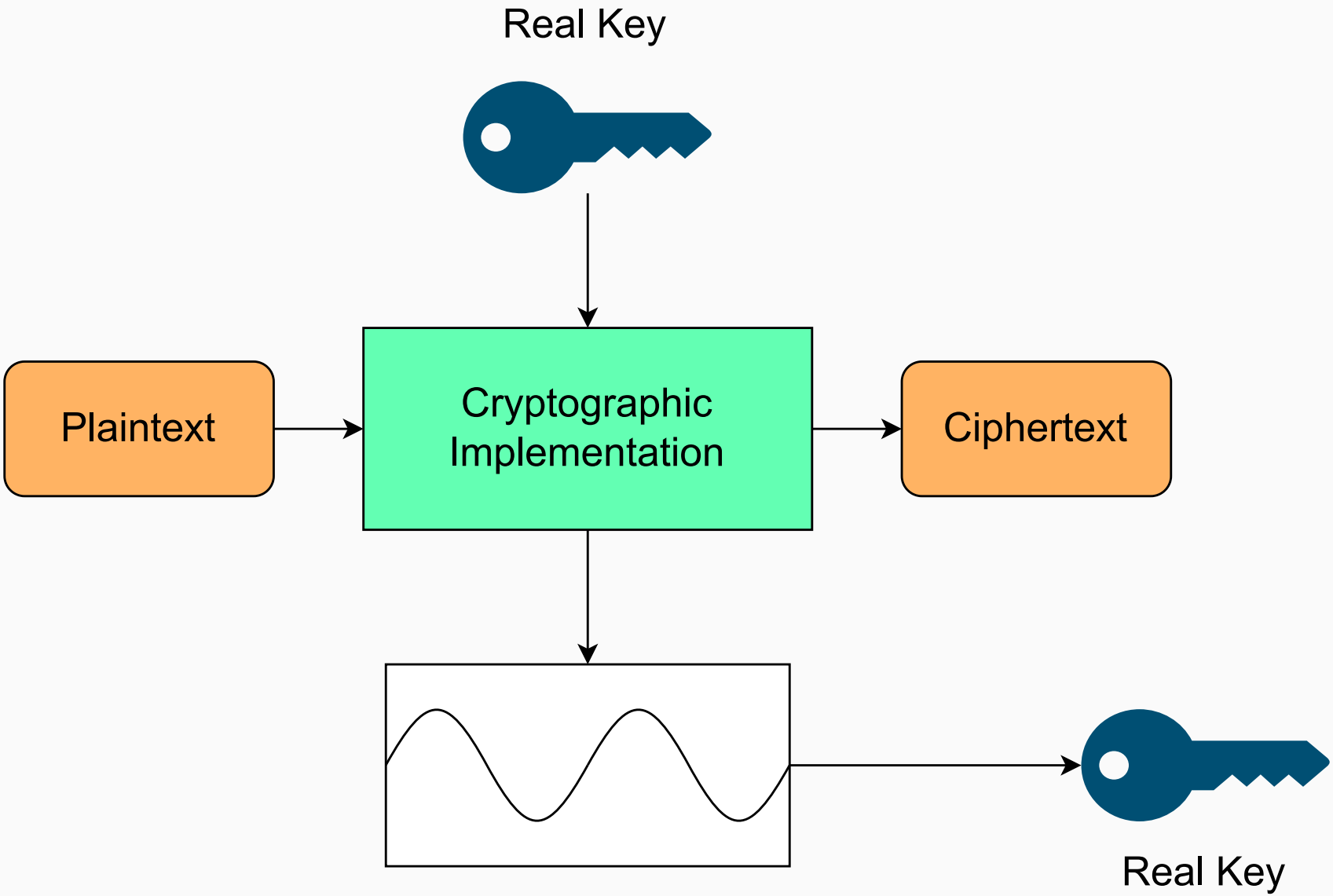
Exploring Leakages within OccPoIs

Exploitation of OccPoIs

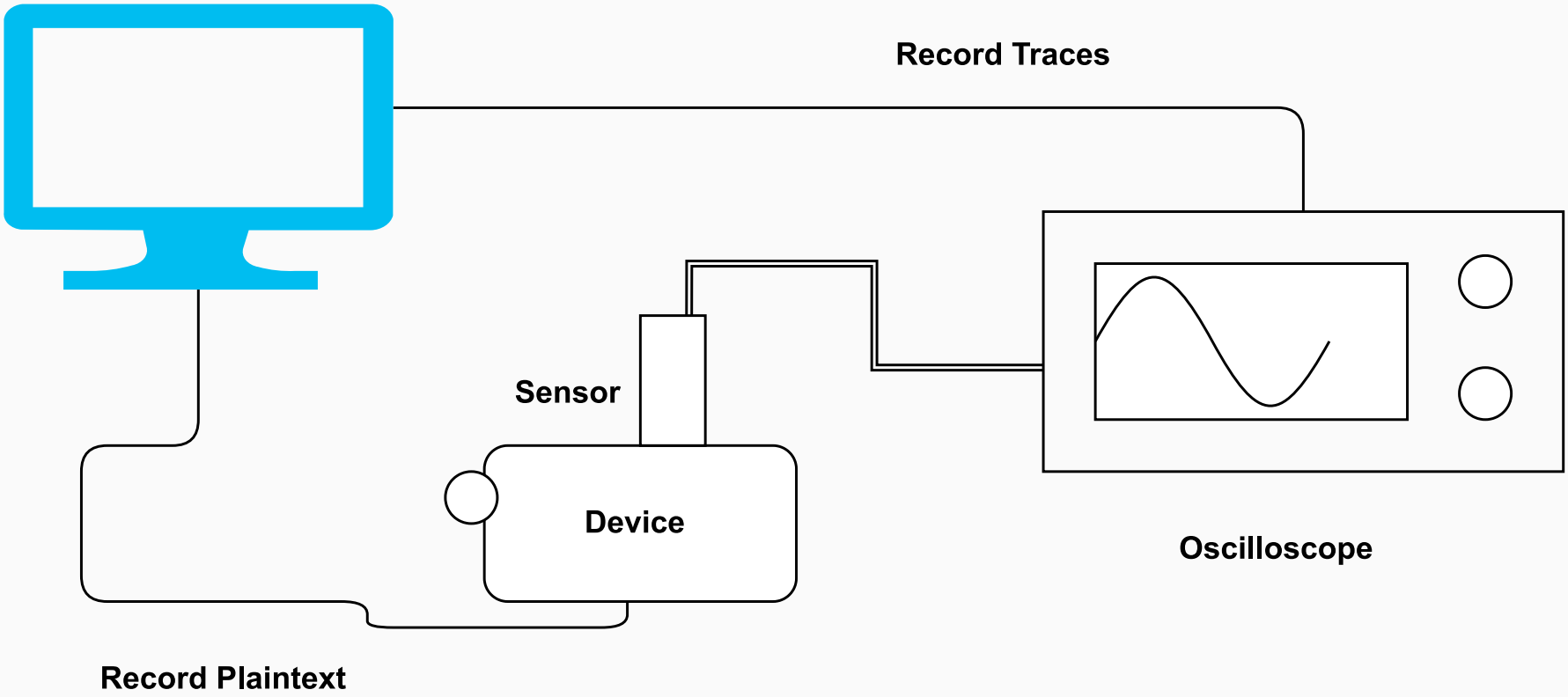
Traces with Desynchronization

Limitation

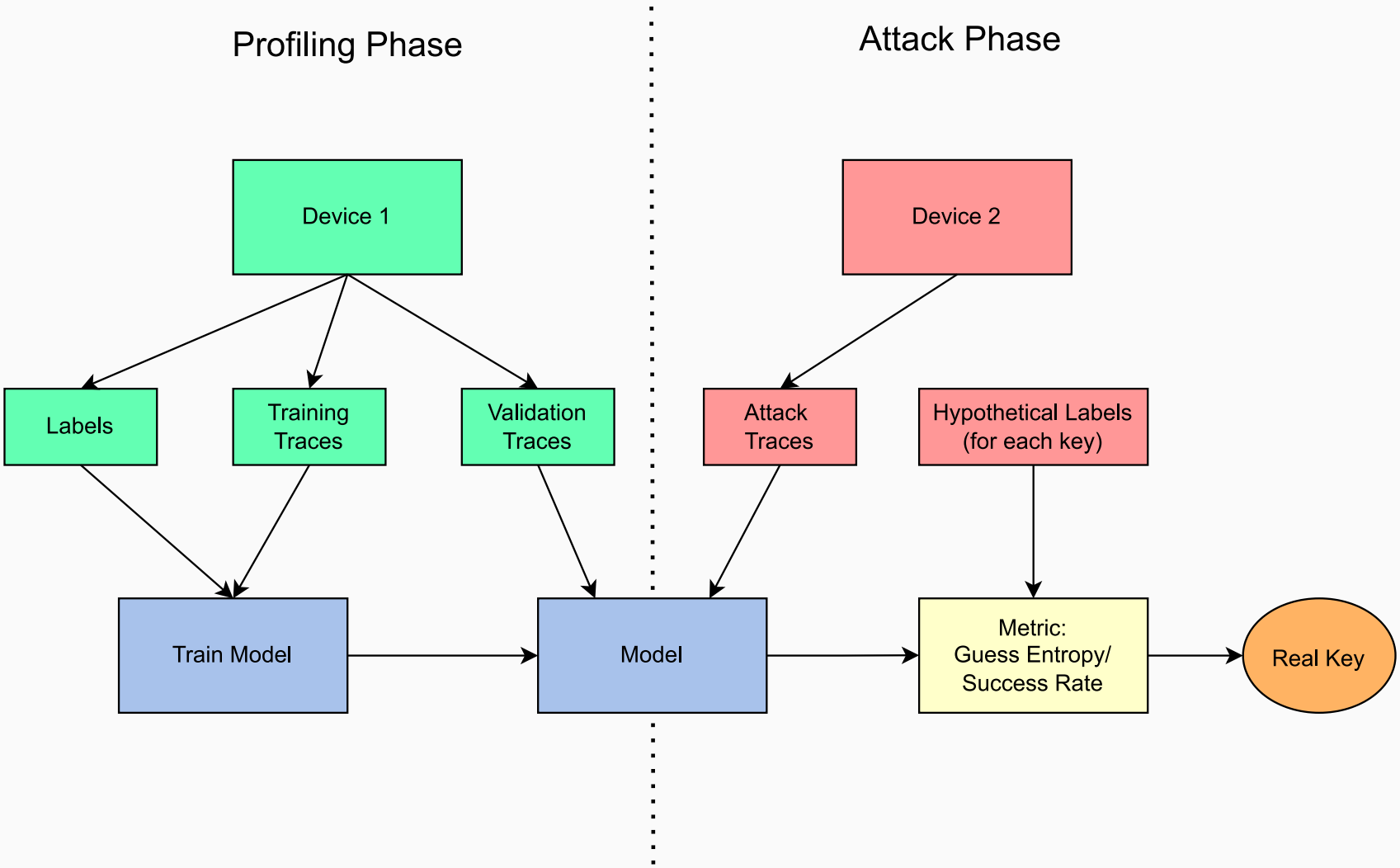
Overview of side channel analysis



Trace acquisition set up

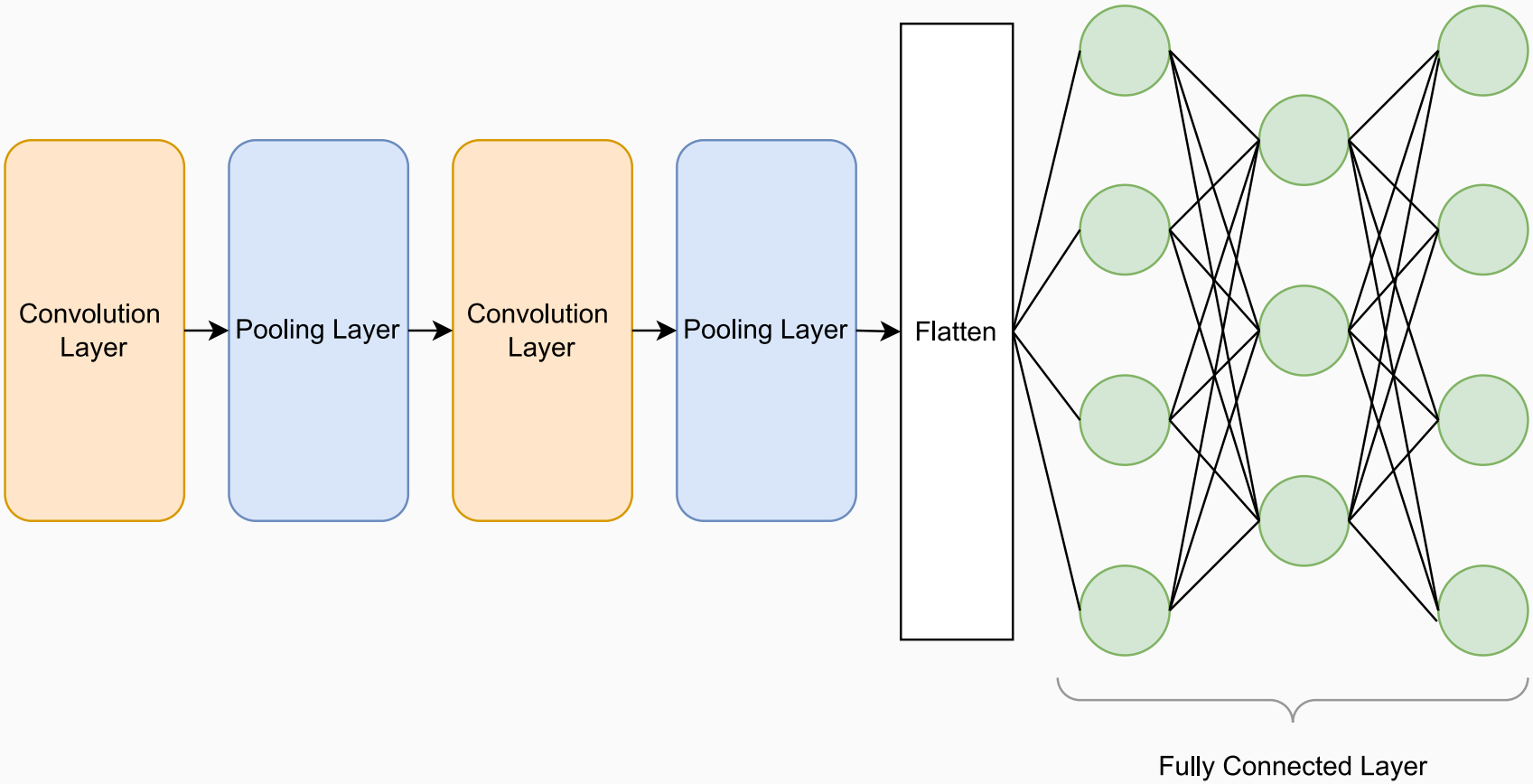


Profiling set up

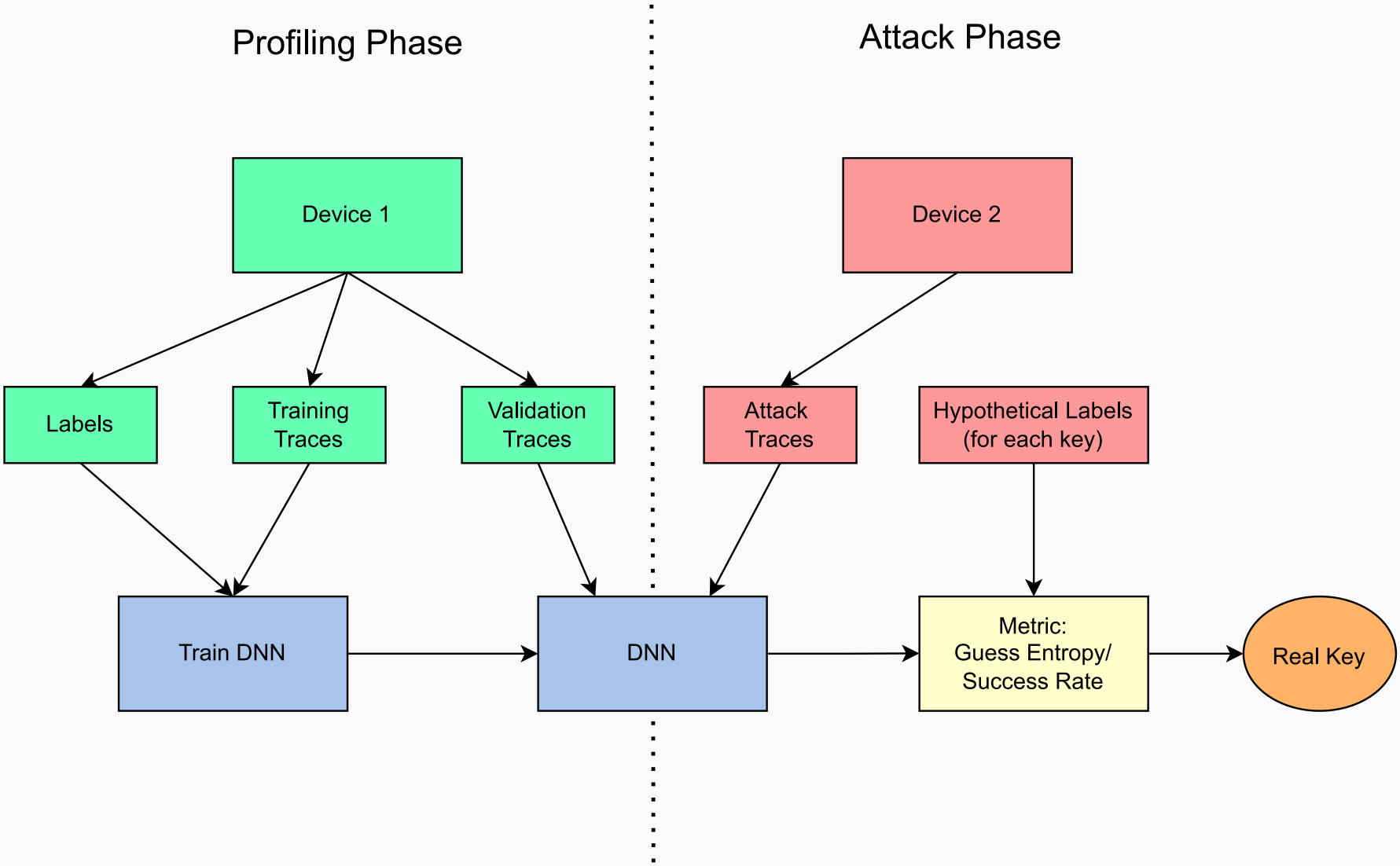


When the Guessing Entropy (GE) is equal 0, it means we have successfully recover the secret key.

Deep neural network (DNN)



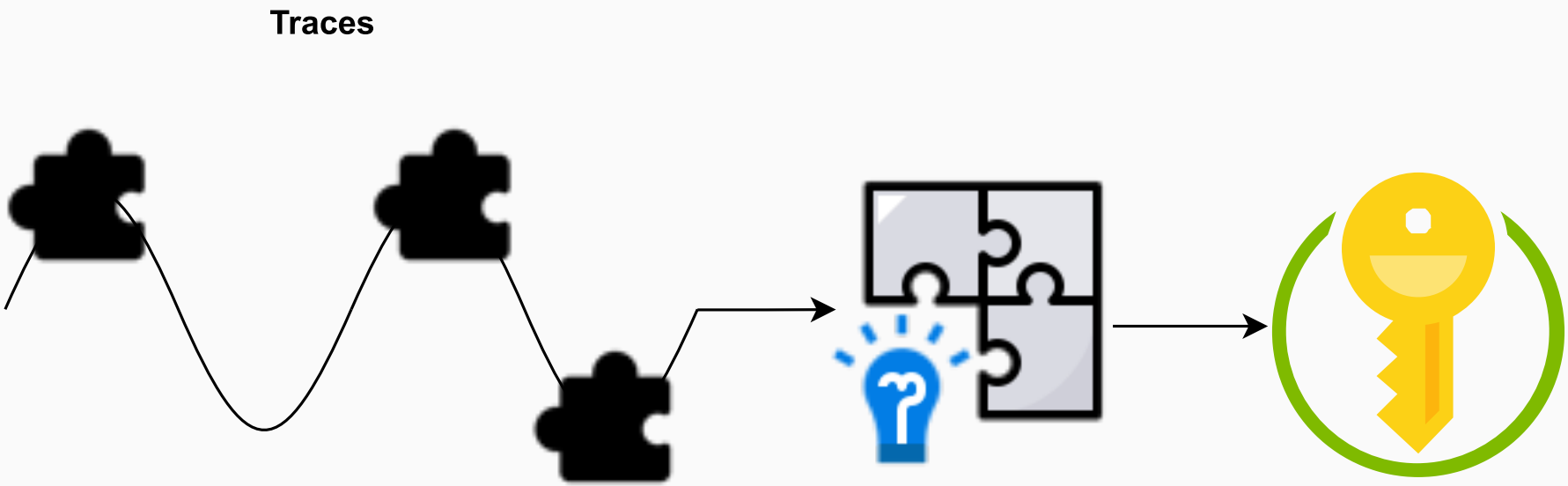
Profiling attack with DNN



Advantage of DNN

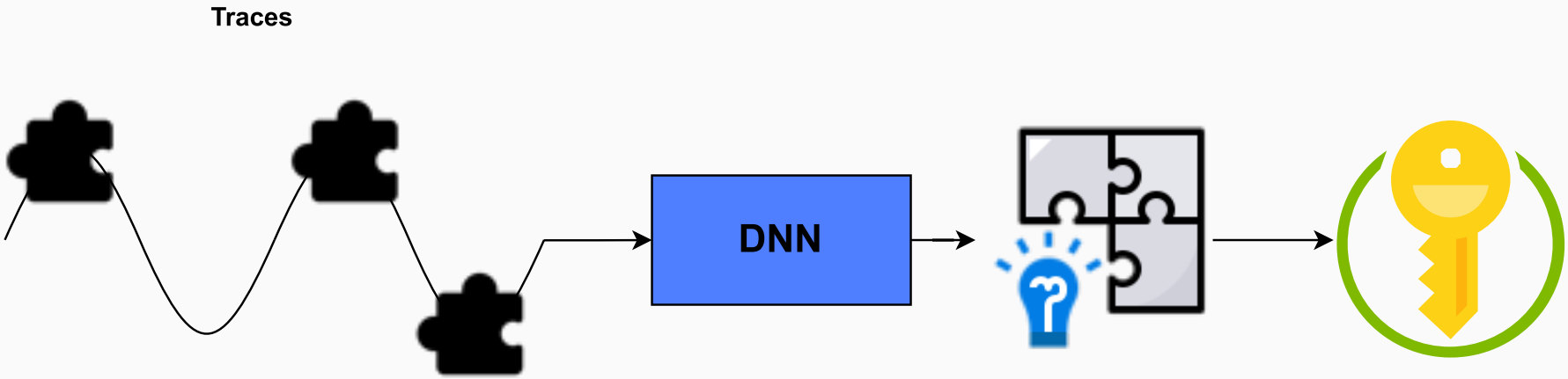
Classical SCA (i.e. Template Attack)	DNN-based attack
More traces to attack	Significantly lesser traces to attack
Preprocessing to remove any desynchronous in the traces	No preprocessing needed
Preprocessing to obtain Points of Interest (PoIs)	No preprocessing needed

Masking



Secret information is split into different shares.

Question



Since the DNN can **combine these shares implicitly** to retrieve the secret key of the target device and **traces contain relevant and redundant sample points**, an inherent question arises:

What features/sample points does a trained DNN use to obtain the secret key?

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$index_{lst}$

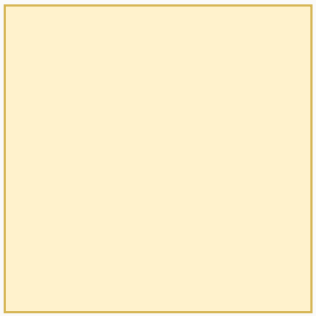
x_0	x_1	x_2
...	...	...
x_{N-3}	x_{N-2}	x_{N-1}

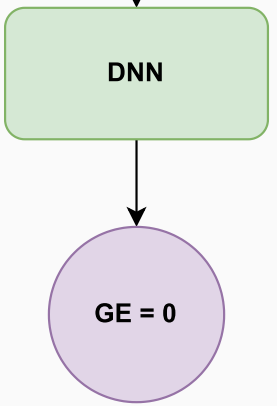
Traces

x_0	x_1	x_2	x_3	x_4	...	...	...	x_{N-3}	x_{N-2}	x_{N-1}
9.04	23.8	31.5	15.1	21.4	...	...	...	61.7	29.8	44.6
14.4	41.0	23.9	16.0	8.9	...	...	...	31.0	29.8	24
...	...	...	...	...	...	...	...	...	...	...
7.52	16.5	31.5	15.1	21.4	...	...	...	61.7	29.8	44.6

Important sample points,

$index_{rev}$





Key Guessing Occlusion (KGO)

$index_{lst}$

x_3	x_0	x_{N-3}
...	...	...
x_{13}	x_{59}	x_1

Traces

x_0	x_1	x_2	x_3	x_4	...	...	...	x_{N-3}	x_{N-2}	x_{N-1}
9.04	23.8	31.5	15.1	21.4	...	...	...	61.7	29.8	44.6
14.4	41.0	23.9	16.0	8.9	...	...	...	31.0	29.8	24
...	...	...	...	...	...	...	...	...	...	...
7.52	16.5	31.5	15.1	21.4	...	...	...	61.7	29.8	44.6

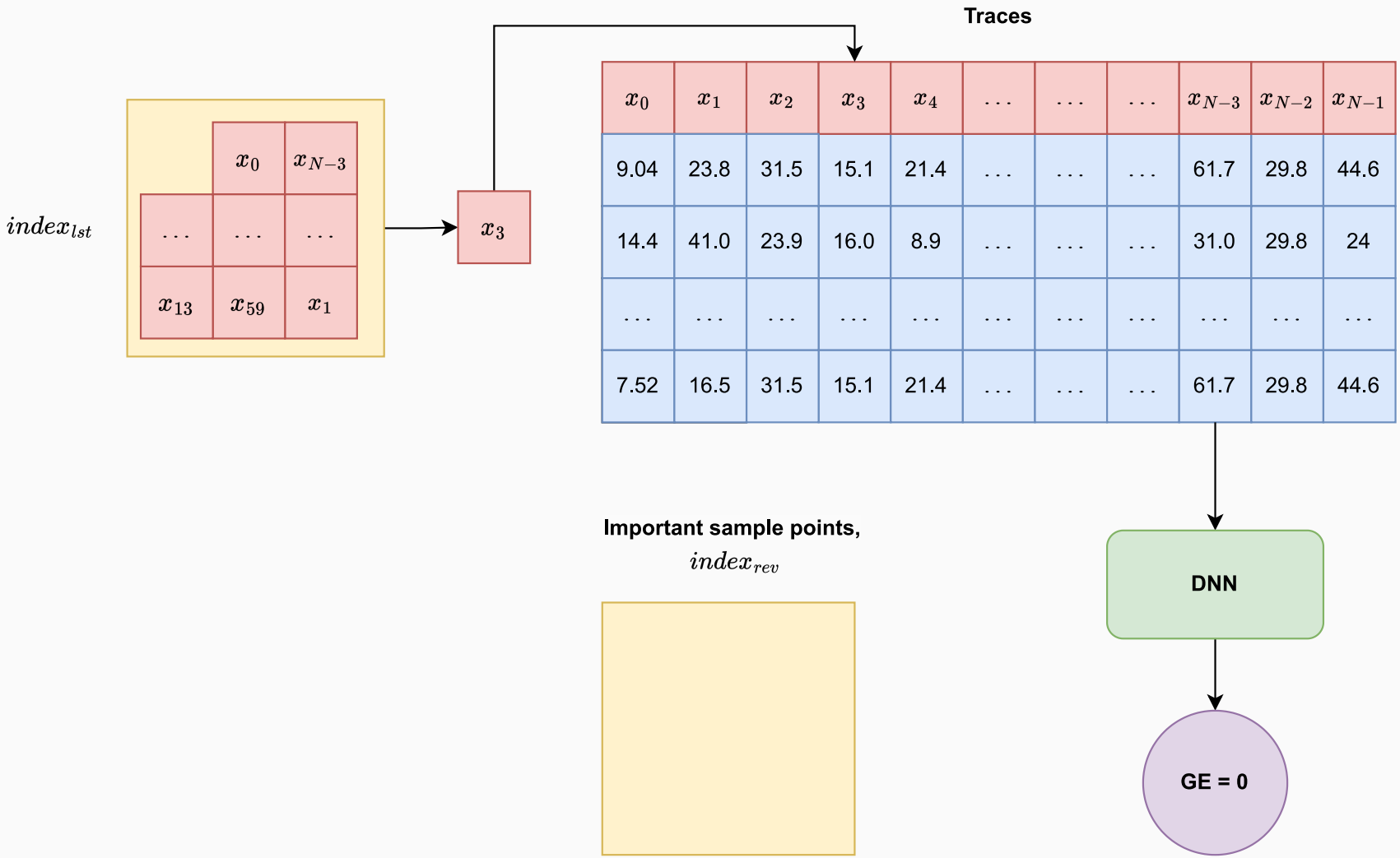
Important sample points,

$index_{rev}$

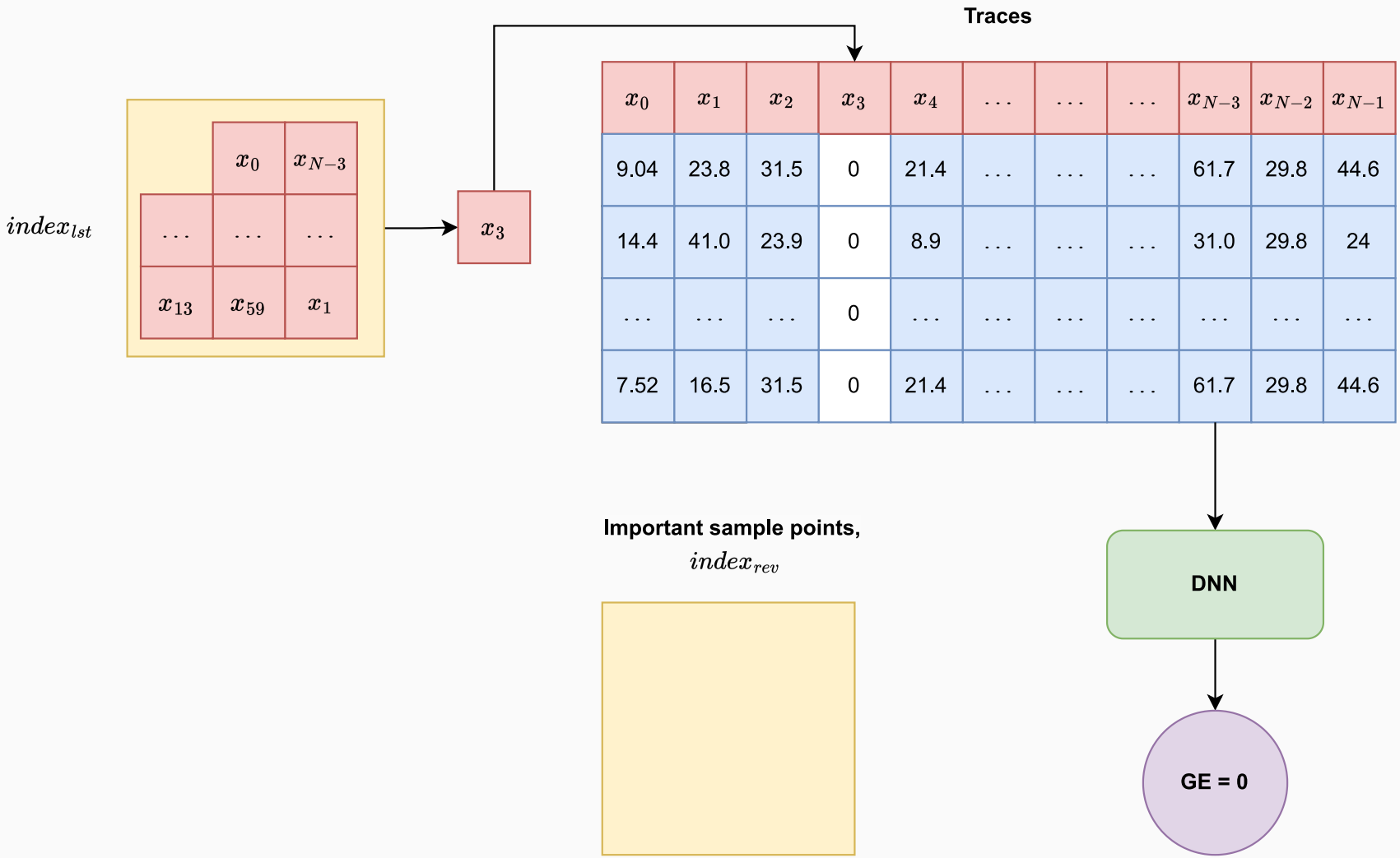
DNN

GE = 0

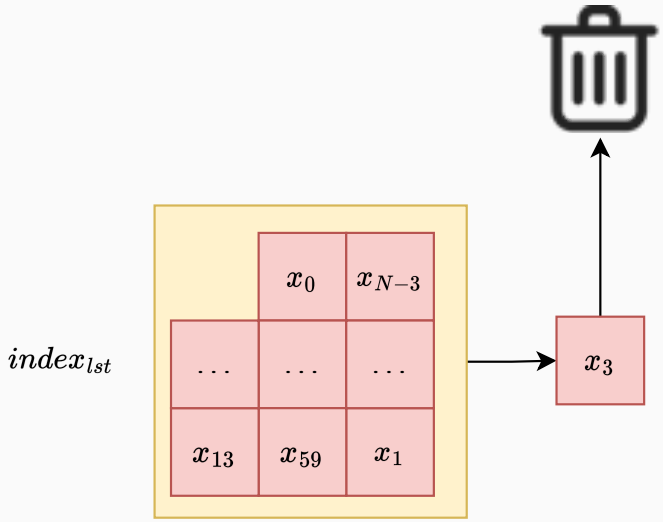
Key Guessing Occlusion (KGO)



Key Guessing Occlusion (KGO)



Key Guessing Occlusion (KGO)



Traces

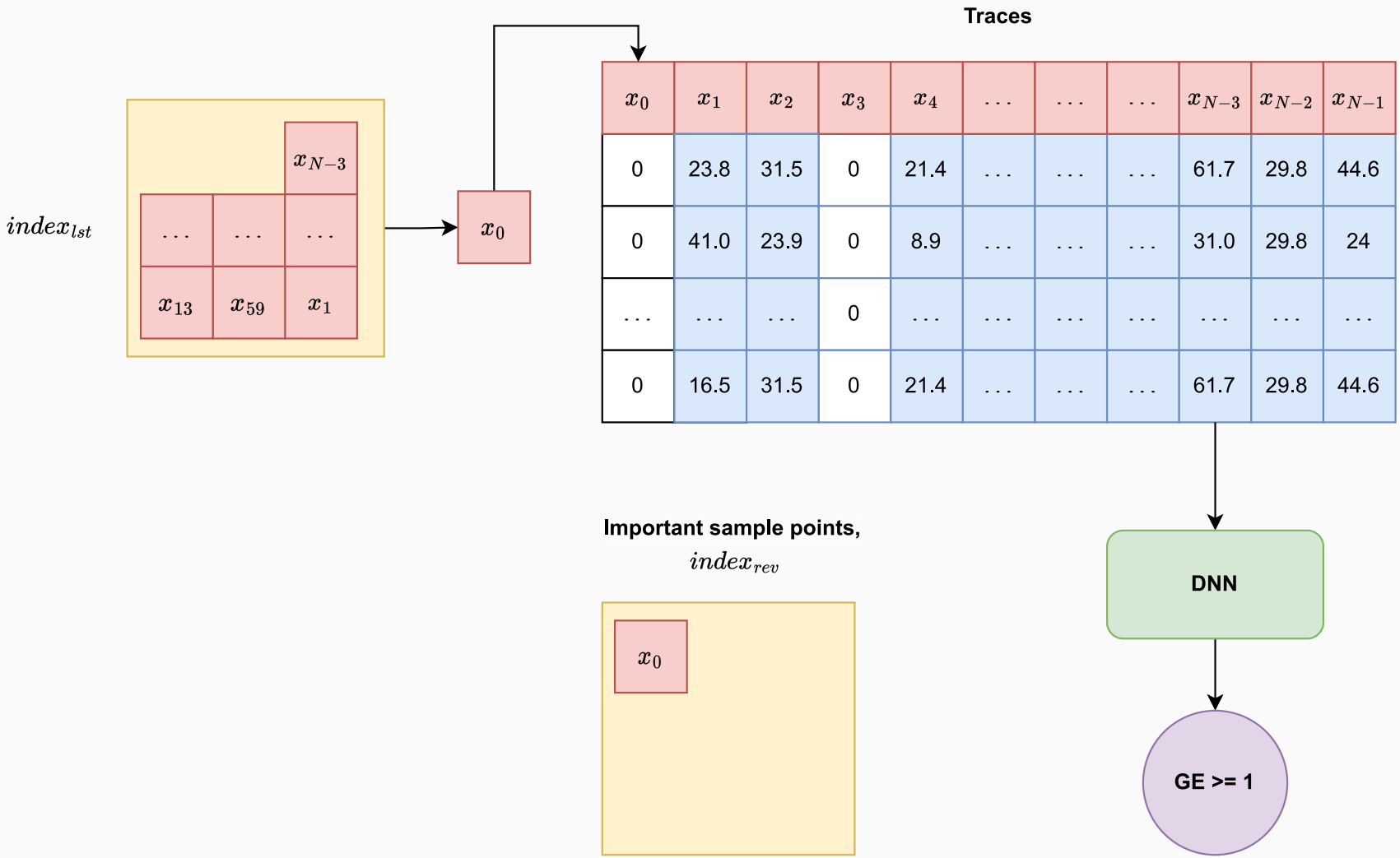
x_0	x_1	x_2	x_3	x_4	...	...	...	x_{N-3}	x_{N-2}	x_{N-1}
9.04	23.8	31.5	0	21.4	...	...	...	61.7	29.8	44.6
14.4	41.0	23.9	0	8.9	...	...	...	31.0	29.8	24
...	...	...	0	...	...	...	...	...	...	...
7.52	16.5	31.5	0	21.4	...	...	...	61.7	29.8	44.6

Important sample points,
$index_{rev}$

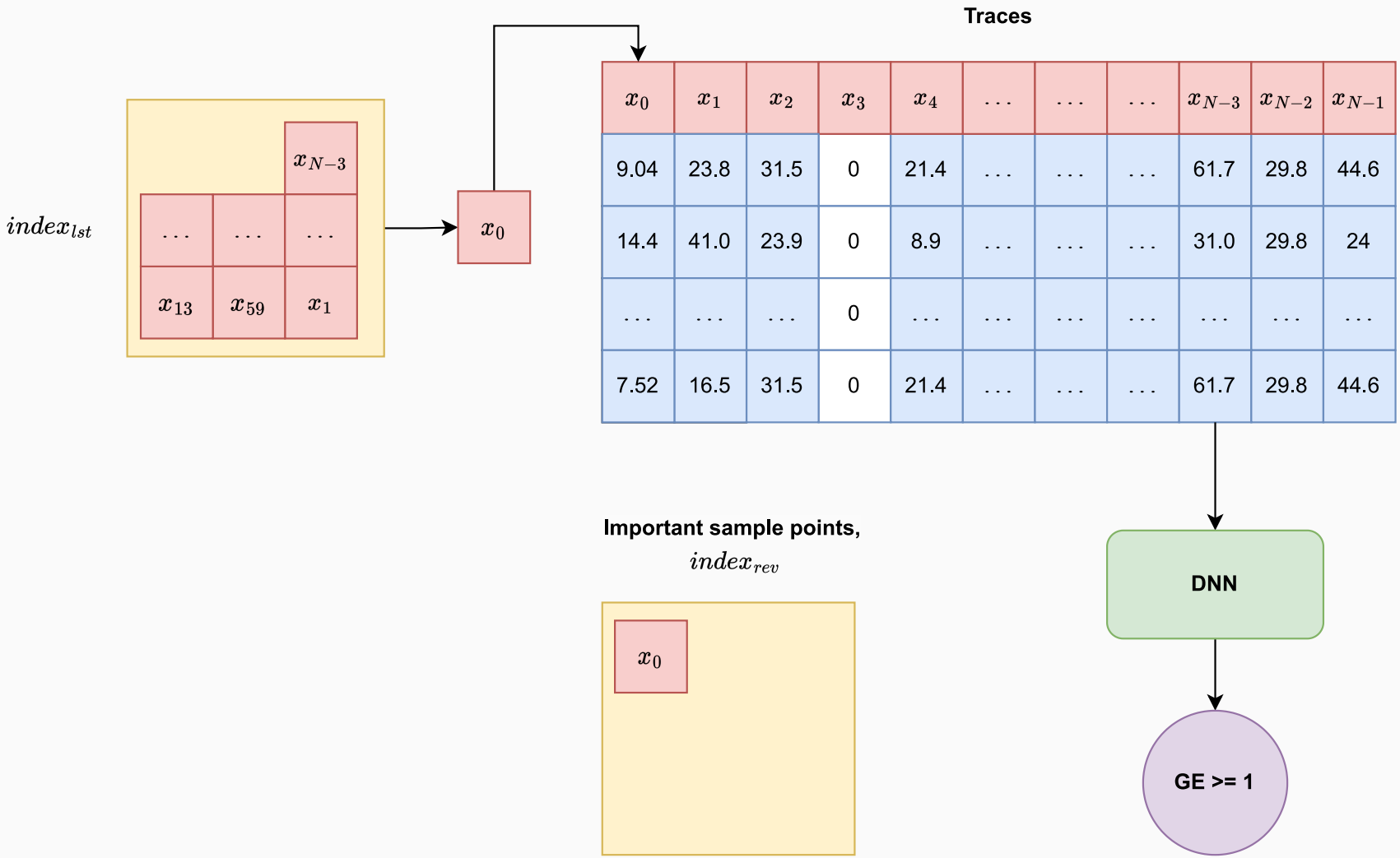
DNN

GE = 0

Key Guessing Occlusion (KGO)

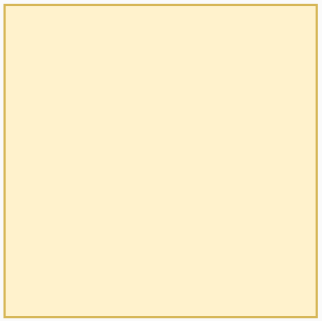


Key Guessing Occlusion (KGO)



Key Guessing Occlusion (KGO)

$index_{lst}$

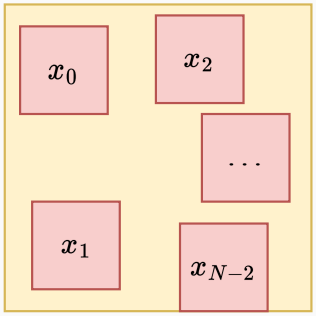


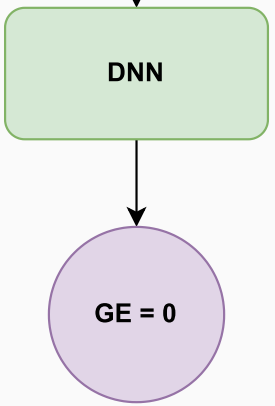
Traces

x_0	x_1	x_2	x_3	x_4	...	...	...	x_{N-3}	x_{N-2}	x_{N-1}
9.04	23.8	31.5	0	0	...	...	...	0	29.8	0
14.4	41.0	23.9	0	0	...	...	...	0	29.8	0
...	...	...	...	...	...	...	...	...	...	...
7.52	16.5	31.5	0	0	...	...	...	0	29.8	0

Important sample points,

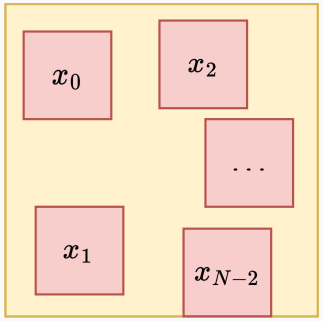
$index_{rev}$





Key Guessing Occlusion (KGO)

$index_{lst}$

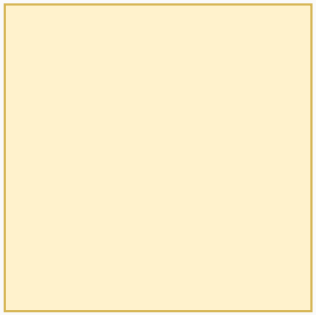


Traces

x_0	x_1	x_2	x_3	x_4	...	...	...	x_{N-3}	x_{N-2}	x_{N-1}
9.04	23.8	31.5	0	0	...	...	...	0	29.8	0
14.4	41.0	23.9	0	0	...	...	...	0	29.8	0
...	...	...	...	...	...	...	...	...	...	...
7.52	16.5	31.5	0	0	...	...	...	0	29.8	0

Important sample points,

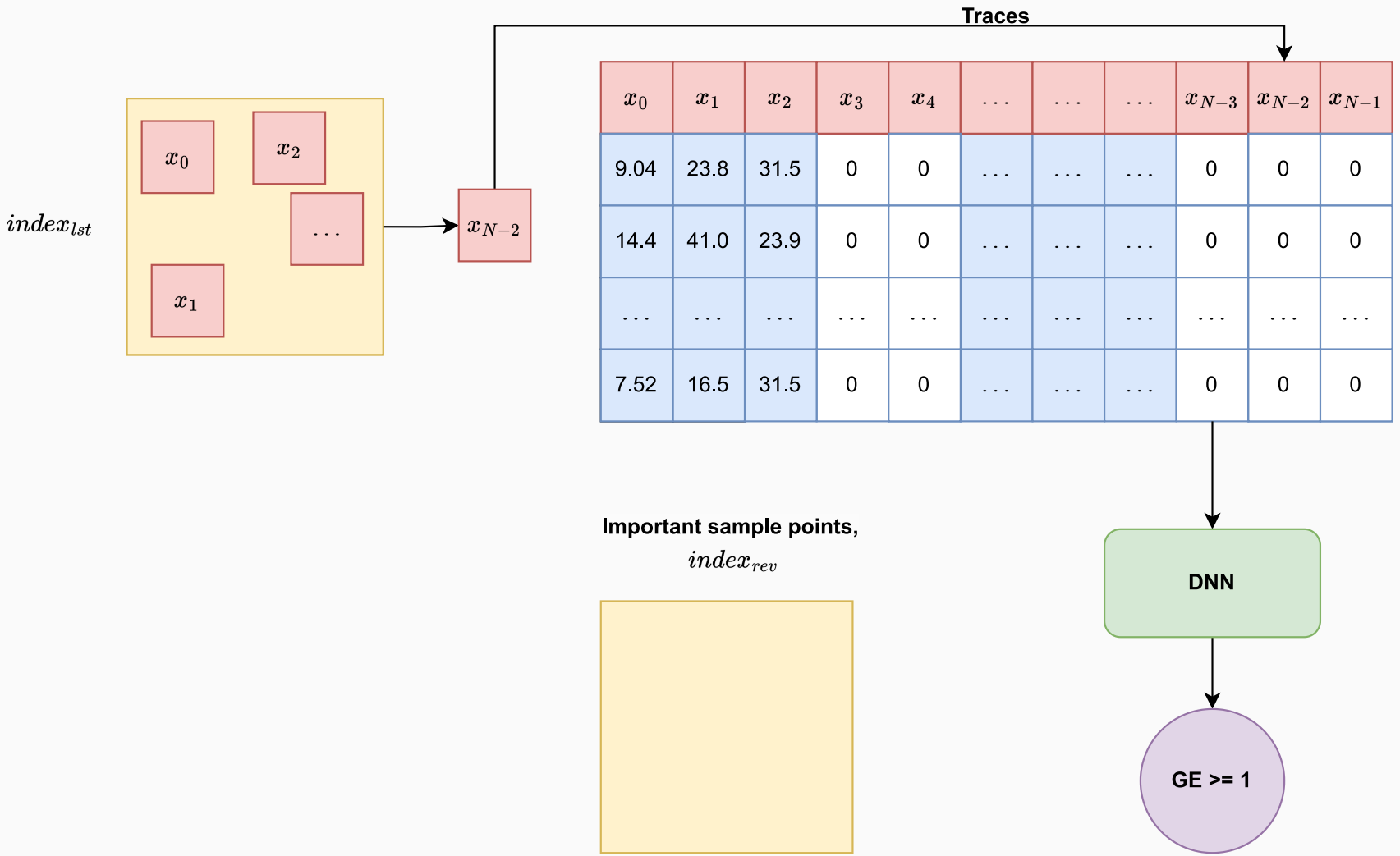
$index_{rev}$



DNN

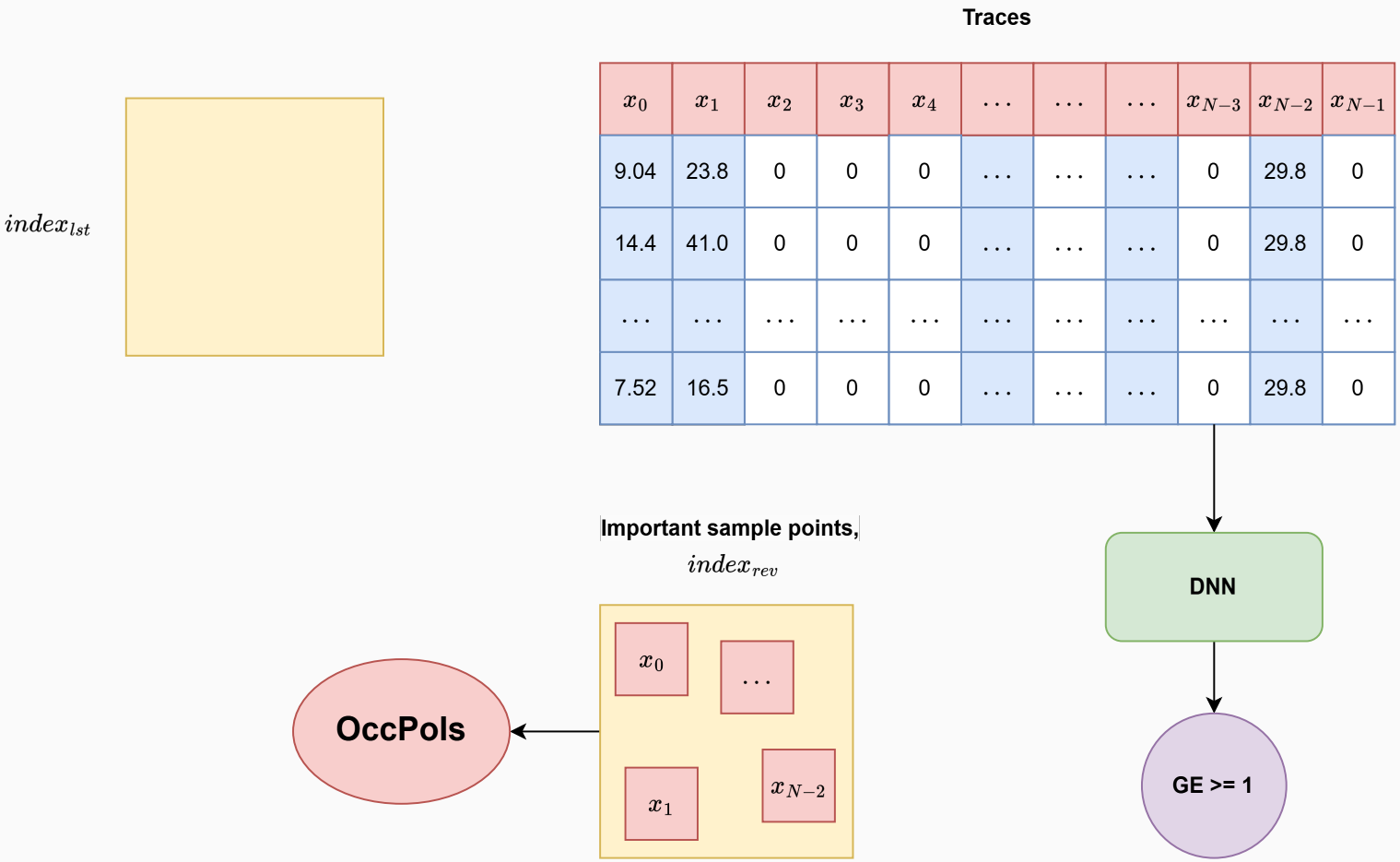
GE = 0

Key Guessing Occlusion (KGO)



Key Guessing Occlusion

We repeat the same until once a sample point is removed the GE will increase.



We call these important sample points as **OccPoIs**.

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Exploring the number of OccPoIs

	CW	ASCADf	ASCADr	AES_HD
Total number of sample points	5000	700	1400	1250
ω	1	5	6	1

Table 1: The number of OccPoIs ω obtained by KGO.

- Relative small number of OccPoIs compare to the total number of samples points.
- In some cases, obtain the least number of sample points required to retrieve the secret key.

Pearson correlation & CPA

Pearson correlation

The Pearson correlation between two vectors $\boldsymbol{x} = (x^{(0)}, \ldots, x^{(N-1)})$ and $\boldsymbol{y} = (y^{(0)}, \ldots, y^{(N-1)})$ is defined as:

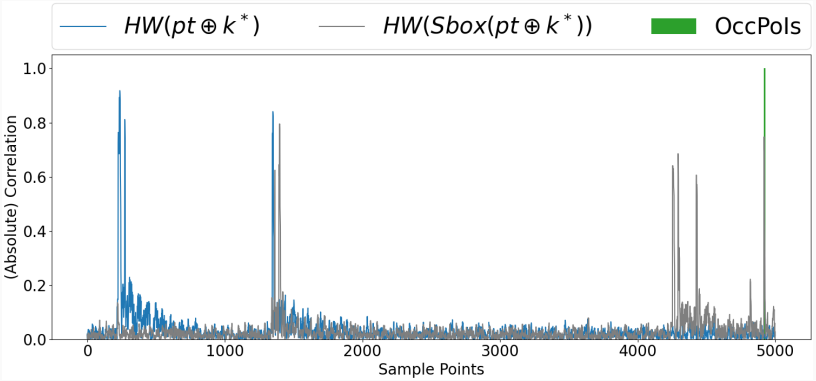
$$r(\boldsymbol{x}, \boldsymbol{y}) = \frac{\sum_{j=0}^{N-1}((x^{(j)} - \bar{\boldsymbol{x}})(y^{(j)} - \bar{\boldsymbol{y}}))}{\sqrt{\sum_{j=0}^{N-1}(x^{(j)} - \bar{\boldsymbol{x}})^2}\sqrt{\sum_{j=1}^{N}(y^{(j)} - \bar{\boldsymbol{y}})^2}} \tag{1}$$

- Measures the degree of linearity between $\boldsymbol{x}$ and $\boldsymbol{y}$.

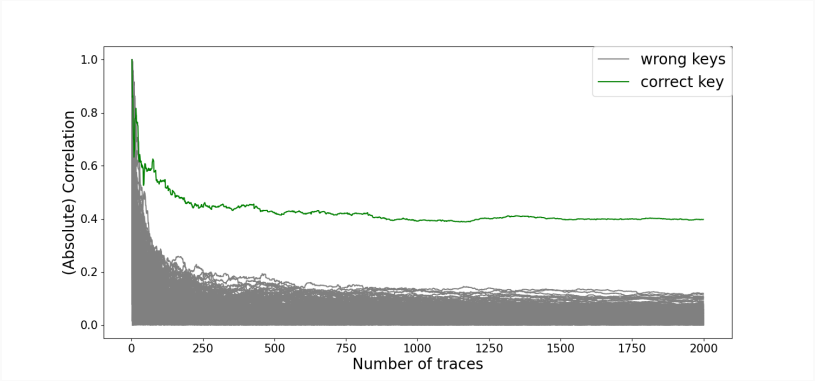
Correlation Power Analysis (CPA)

- For each key guess k, compute r between every sample point of all traces and the hypothetical leakage Z_k (e.g. $Sbox(pt \oplus k)$).
- Let ρ_k be the largest $|r|$ over all the sample points. The key k^* with the largest ρ_{k^*} is considered the correct key.

Chipwhisperer dataset



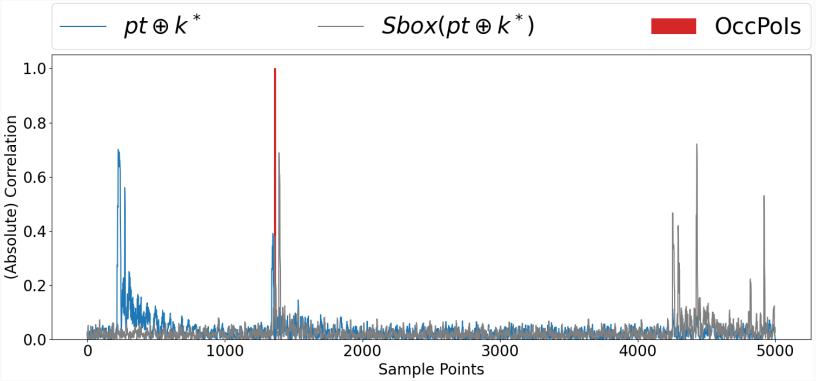
(a) CPA with respect to sample points.



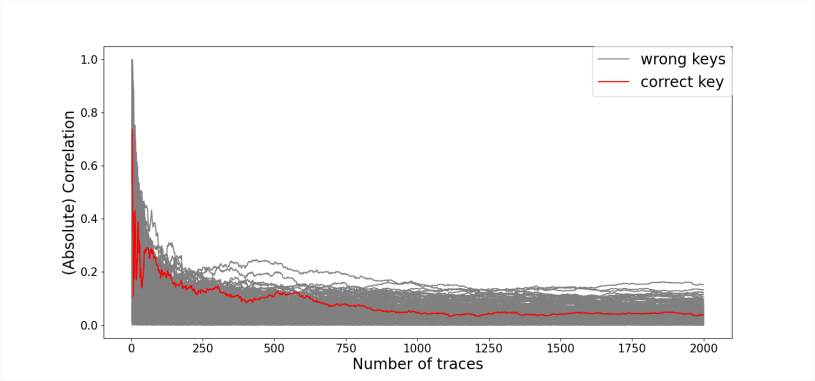
(b) CPA on the OccPoI with respect to the number of traces.

Figure 1: CPA in CW (HW).

Chipwhisperer dataset



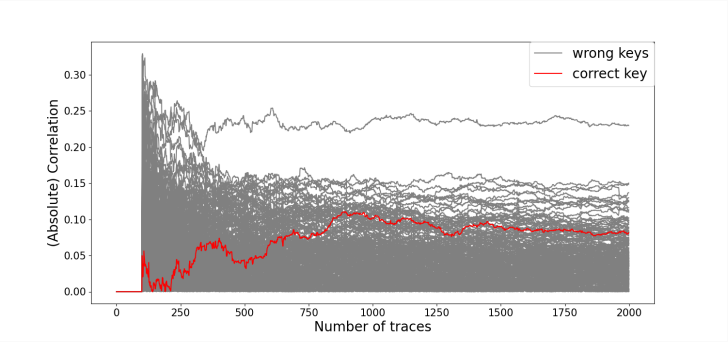
(a) CPA with respect to sample points.



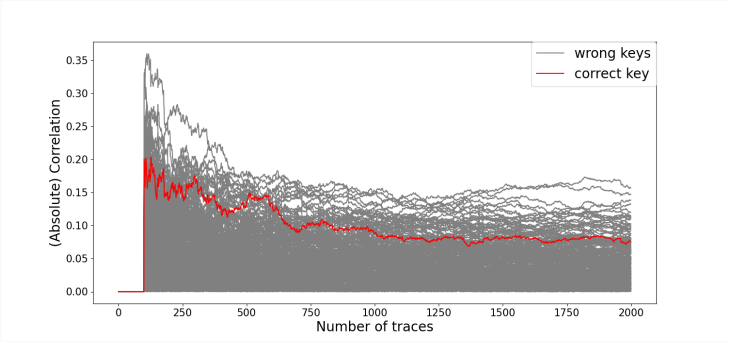
(b) CPA on the OccPoI with respect to the number of traces.

Figure 2: CPA in CW (ID).

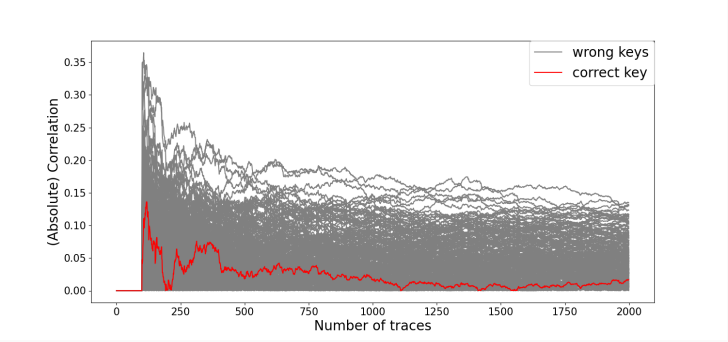
Chipwhisperer dataset



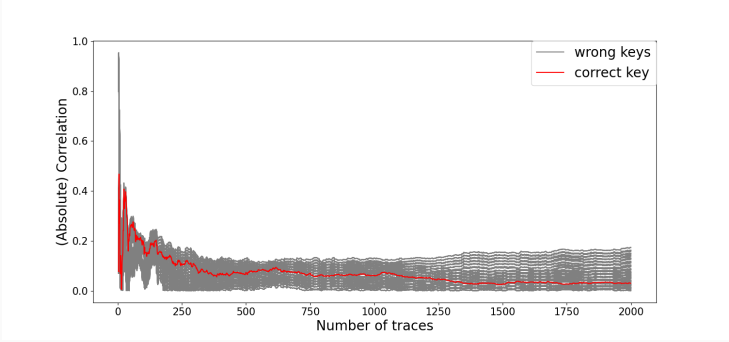
(a) $HW(Sbox(pt \oplus k^*))$.



(b) $MSB(Sbox(pt \oplus k^*))$.



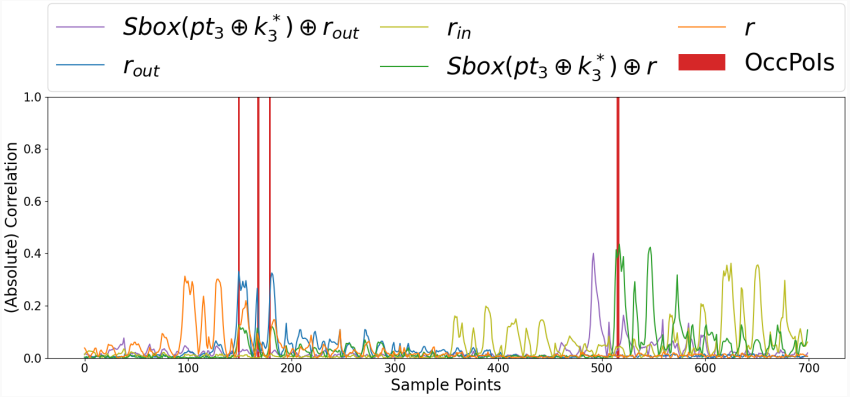
(c) $LSB(Sbox(pt \oplus k^*))$.



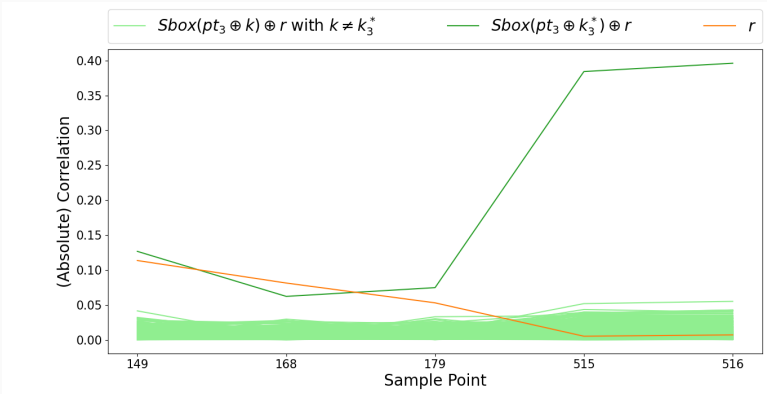
(d) $pt \oplus k^*$.

Figure 3: CPA with other leakage model with respect to the number of traces in CW (ID).

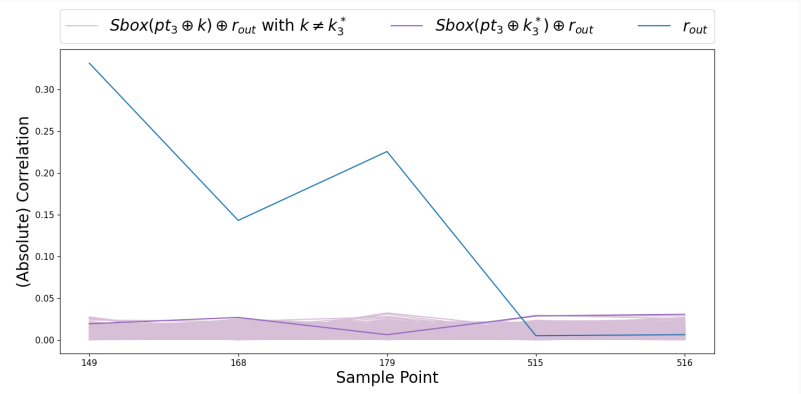
ASCADf dataset



(a) CPA with respect to sample points for ASCADf.

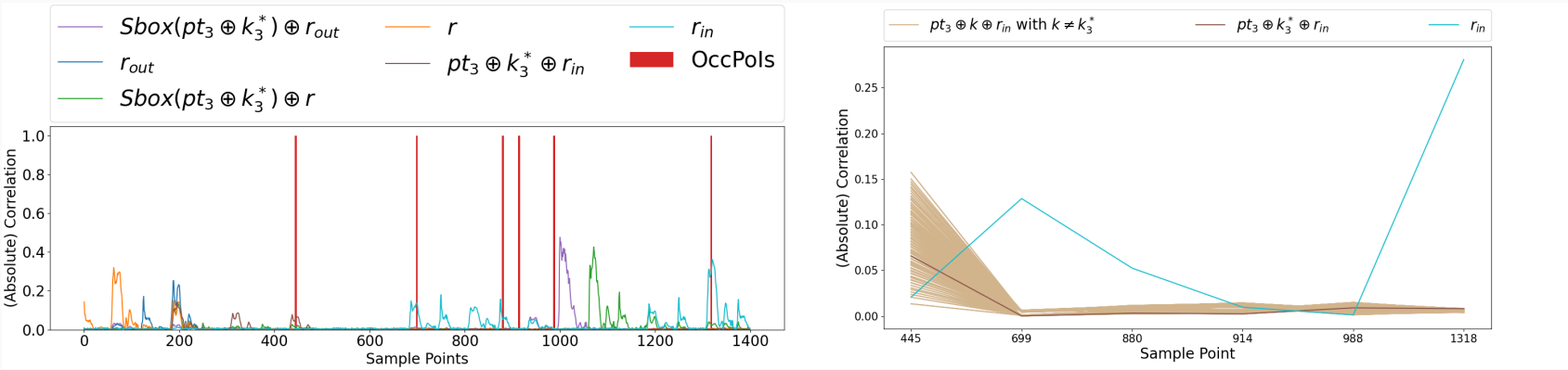


(b) CPA of $Sbox(pt_3 \oplus k_3^*) \oplus r$ and r on KGO's points for ASCADf.



(c) CPA of $Sbox(pt_3 \oplus k_3^*) \oplus r_{out}$ and r_{out} on KGO's points for ASCADf.

ASCADr dataset



(a) CPA with respect to sample points. **(b)** CPA on KGO's points.

Figure 5: CPA in ASCADr.

- It learns a the leakage $pt_3 \oplus k_3^* \oplus r_{in}$ and r_{in} instead.
- There exist sample points which are not highly correlated to any of the leakage model tested (e.g., 988 sample point).
- DNN learns complex information from these sample points that first-order CPA missed.

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Feature selection tool for TA

- It has been demonstrated that extracting relevant points, can lead to significantly more effective attacks in Template Attack (TA).
- Since the number of OccPoIs is **much smaller** than the total number of sample points and DNN **could recover the secret key** from these sample points, we use KGO as a **feature selection tool**.

Exploitation of OccPoIs as feature selection tool

We compared with the classical features selection and other explainability techniques for Template Attack (TA).

	CW	ASCADf	ASCADr	AES_HD
SOSD	**3**	$> 10k$ ($GE = 2$)	$> 100k$ ($GE = 103$)	2623
SOST	9	$> 10k$ ($GE = 68$)	$> 100k$ ($GE = 51$)	2718
CPA (first-order)	11	$> 10k$ ($GE = 67$)	$> 100k$ ($GE = 162$)	2809
CPA (multi.)	-	367	**7184**	-
Saliency	9	$> 10k$ ($GE = 248$)	$> 100k$ ($GE = 210$)	3515
LRP	8	$> 10k$ ($GE = 100$)	50061	**2265**
1-Occlusion	18	$> 10k$ ($GE = 162$)	$> 100k$ ($GE = 7$)	3381
KGO	10	**313**	42991	6176

Table 2: NT_{GE} of the TA when using various feature selection techniques.

- KGO able to obtain $GE = 0$ for all datasets.
- KGO obtain better results than second-order CPA for ASCADf.
- KGO is stable compare to other techniques.

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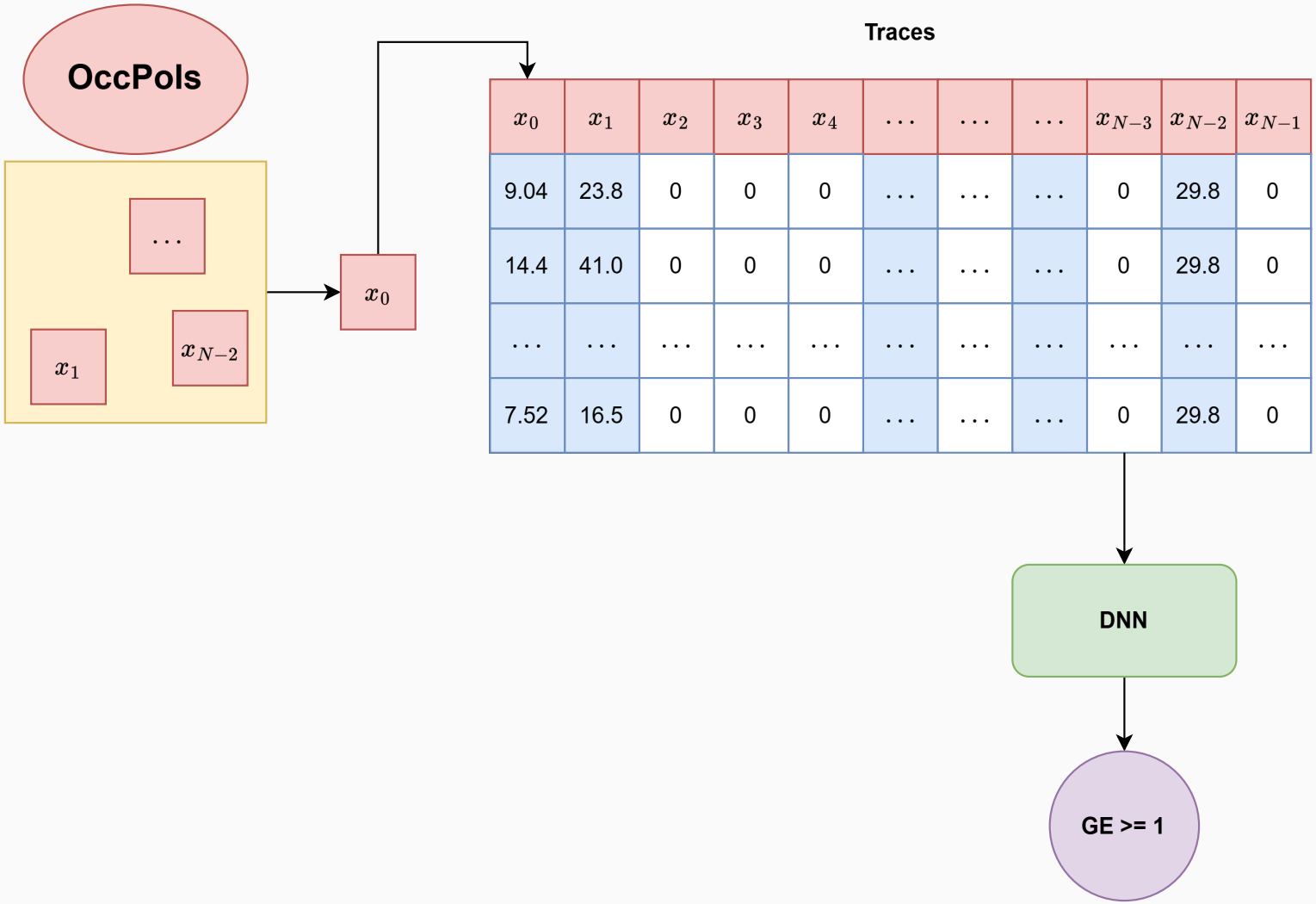
Traces with Desynchronization

Limitation

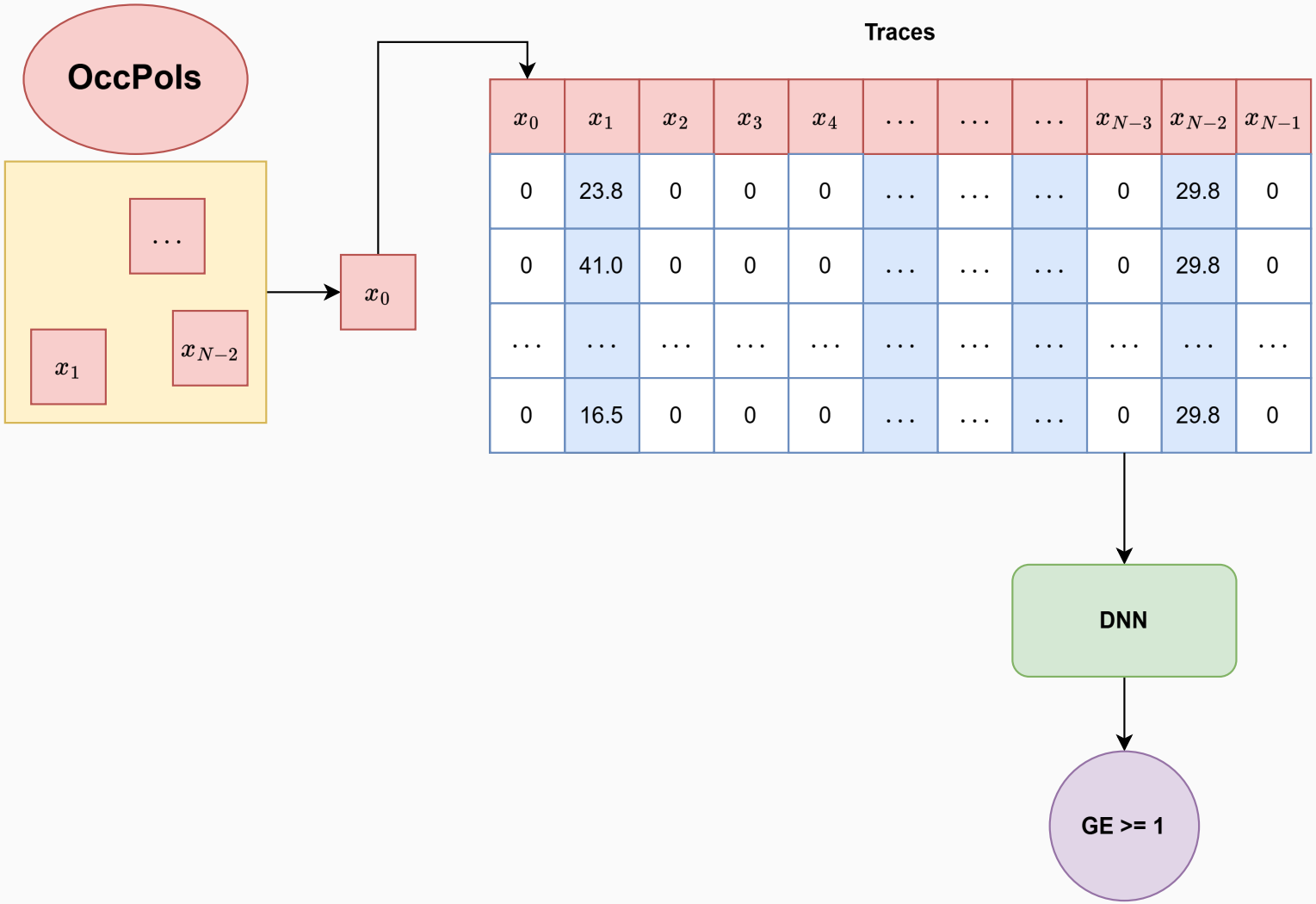
Traces with Desynchronization

- Currently, to know which sample points are leaking for desynchronized traces, one could resynchronize them and apply CPA or Signal-to-Noise ratio (SNR) to observe any leakage.
- For unprotected datasets, one could increase the number of traces to see identify the PoIs with CPA/SNR.
- Identifying the PoIs of traces from implementations that are **protected by both desynchronization and masking**, it is **not possible without resynchronizing** the trace.
- The process of **resynchronizing the traces is tedious.**

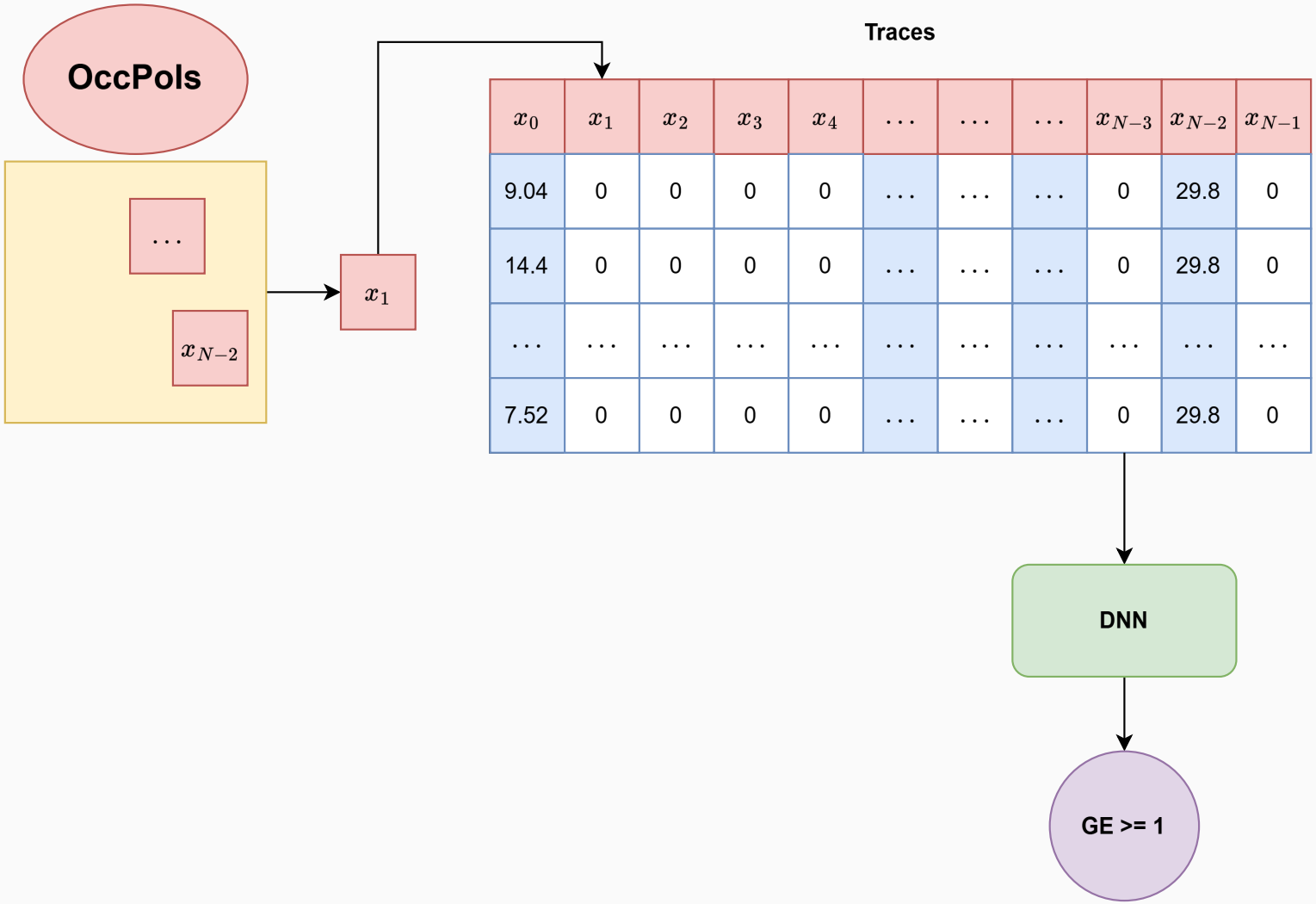
Visualizing leakages 1-KGO



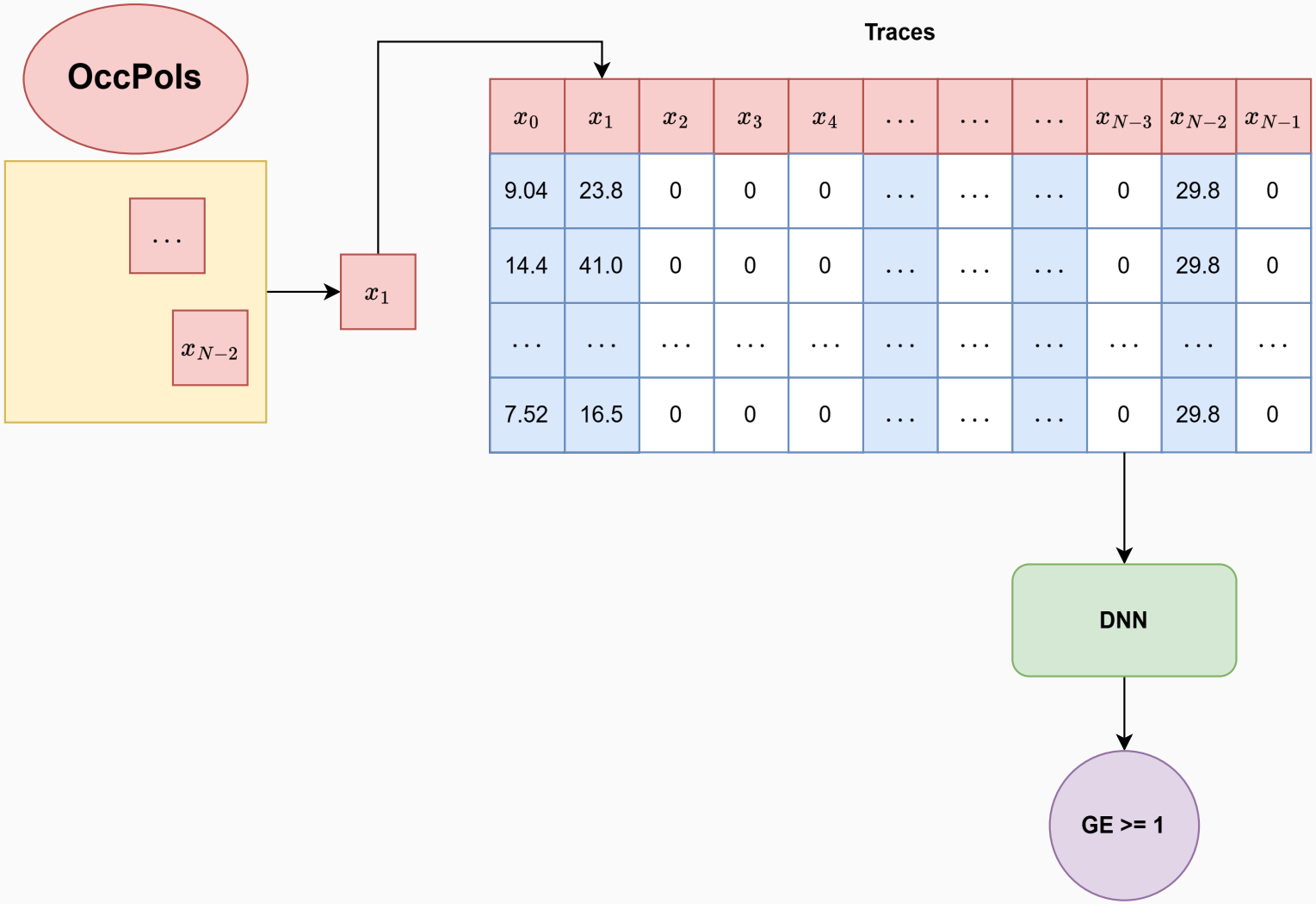
Visualizing leakages 1-KGO



Visualizing leakages 1-KGO

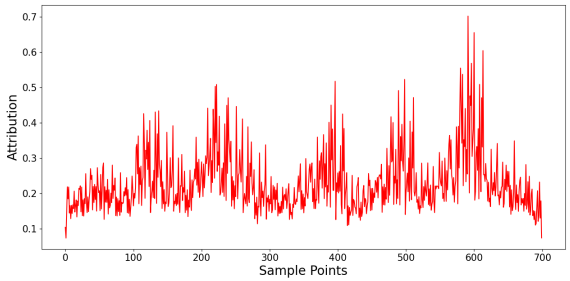


Visualizing leakages 1-KGO

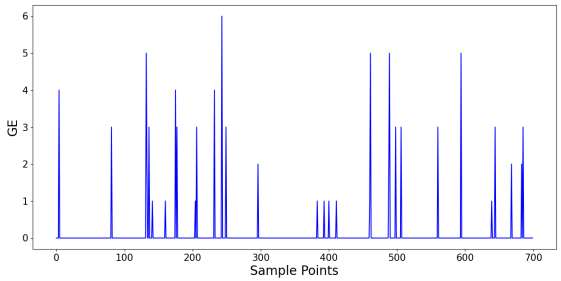


The larger the *GE*, the more contribution the sample point is towards the recovery of key.

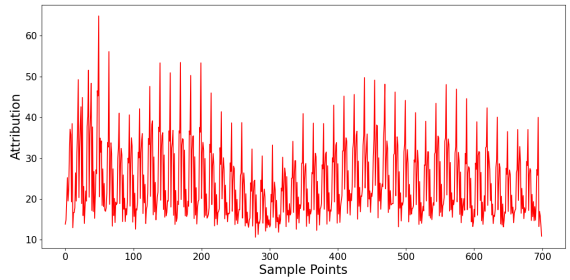
Traces with Desynchronization



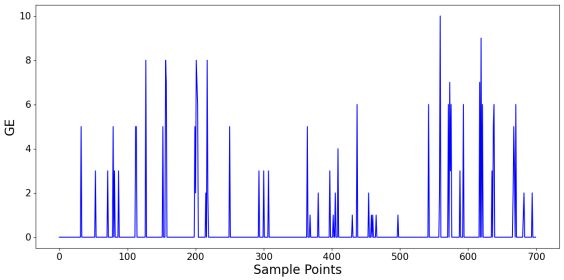
(a) Saliency Map in ASCADf_desync50.



(b) 1-KGO of DNN in ASCADf_desync50.



(c) Saliency Map in ASCADf_desync100.



(d) 1-KGO of DNN in ASCADf_desync100

Figure 6: Explainability of DNN in ASCADf with desynchronization.

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Introduction

Key Guessing Occlusion (KGO)

Exploring Leakages within OccPoIs

Exploitation of OccPoIs

Traces with Desynchronization

Limitation

Limitation

- Trade-off in terms of time for interpretation.

	CW	ASCADf	ASCADr	AES_HD
Time Taken (hrs)	26.3	21.3	227.7	44.9

Table 3: Time taken to run KGO.

- KGO does not allow to choose the number of sample points.
- If KGO does not consider the sample points, it does not mean they are not leaking. There might be more than one set of OccPoIs.

Thank You!